- Castner Kellner process is used for preparation of caustic soda.
- Bleaching powder is prepared by the reaction of lime water with chlorine gas.
- The order of acidic strength of hydrogen halides is HI > HBr > HCl > HF.
- Chlorine acts as auto oxidizing and reducing agent in certain chemical reactions.
- Sulphuric acid is a good dehydrating agent.
- Sulphuric acid reacts with metals and non metals, it acts as an oxidizing agent.
- Similarities in the properties exhibited by certain pairs of elements among the groups of representative elements are known as diagonal relationship.

Multiple Choice Questions

(i) Melting and boiling points regularly decrease down the group in:
(a) Group IA (b) Group IIIA
(c) Group VIIA (d) Group VIIIA

(ii) The s-block element forms super oxide when burned in air is:
(a) Li (b) Na
(c) K (d) Mg

(iii) The general formula of nitrides for alkaline earth metal is:
(a) MN_3 (b) M_2N_3
(c) M_3N (d) M_3N_2

(iv) The alkali metal that gives a yellow color during a flame test is:
(a) Na (b) K
(c) Rb (d) Cs

(v) The chemical commonly used in fireworks is:
(a) Sodium bicarbonate (b) Bleaching powder
(c) Potassium nitrate (d) Potash Alum

(vi) Cathode in Castner Kellner cell is:
(a) Titanium blocks (b) Carbon rods
(c) Mercury (d) Iron container

(vii) The diagonal member of beryllium is:

(a) Mg (b) Al

(c) Si (d) C

(viii) The catalyst commonly used in the conversion of sulfur dioxide to sulfur trioxide during the preparation of sulphuric acid is:

(a) Vanadium pentoxide (V_2O_5) (b) Copper (Cu)

(c) Iron (Fe) (d) Nickel (Ni)

(ix) Oil of vitriol refers to:

(a) Borax (b) Sulphuric acid

(c) Alum (d) Caustic soda

(x) The best oxidizing agent among halogens is:

(a) F_2 (b) Cl_2

(c) Br_2 (d) I_2

Short Questions

1. Give reasons for the following:

(i) Ionization energy decreases from top to bottom in s-block elements?

(ii) Boiling point of halogens increase down the group in the periodic table?

(iii) Gallium has smaller atomic radii than aluminum despite being below the aluminum in group IIIA?

(iv) Electronegativities of alkali metals decrease from Li to Cs?

(v) Alkali metals are good conductor of electricity?

(vi) Acidity of hydrogen halides increase from HF to HI?

(vii) Fluorine is the strongest oxidizing agent?

2. What is flame test? Mention the colour of flame of alkali metals.
3. Explain the auto oxidizing and reducing properties of chlorine.
4. What is meant by a diagonal relationship? Mention three pairs of representative elements that show diagonal relationship.
5. Discuss the group trend of Ionization energy in group IIIA of periodic table.
6. Write down three properties of beryllium that show its unique behavior in group IIA.

- Coordination number is the number of coordinate bonds formed between ligands and the transition metal in a complex molecule or ion.
- The complexes of multidentate ligands are known as chelates.
- Potassium dichromate and potassium permagnate are powerful oxidizing agents.
- Transition elements have ability to form alloys with each other due to the similarities in their atomic sizes.
- Stainless steel is the most common alloy of iron. It is made up of iron, chromium and nickel.
- The most common ore of copper is chalcopyrite ($CuFeS_2$).
- Refining of copper is carried out in an electrolytic tank where thick plates of blister copper serve as anode and thin sheets of pure copper act as cathode.

Multiple Choice Questions

(i) Zn^{+2} ion is colourless because:
(a) Its undergoes d-d transition of electron
(b) Its 3d orbitals have all unpaired electrons
(c) Its 3d orbitals have all paired electrons
(d) Its d orbitals split up into t_2g and eg

(ii) The coordination number of cobalt in $Na_4[Co(C_2O_4)_3]$ is:
(a) 3 (b) 4
(c) 6 (d) 7

(iii) An example of a bidentate ligand among the following is:
(a) $O\bar{H}$ (b) $C_2O_4^{-2}$
(c) Cl^- (d) $C\bar{N}$

(iv) A highly paramagnetic ion among the following is:
(a) Fe^{+2} (b) Fe^{+3}
(c) Co^{+2} (d) Cr^{+3}

(v) The highest oxidation state of chromium is:
(a) +4 (b) +5
(c) +6 (d) +7

(vi) The element is not used for electroplating:
(a) Zinc (b) Tin
(c) Chromium (d) Manganese

(vii) The steel is typically used in making Fry pans:
(a) Carbon steel (b) Stainless steel
(c) Tool Steel (d) Alloy Steel

(viii) The step which involved in the extraction of copper from chalcopyrite ore in the elimination of gangue impurities is:
(a) Concentration (b) Roasting
(c) Smelting (d) Bessemerization

(ix) 5d series of outer transition elements is:
(a) Sc to Zn (b) Y to Cd
(c) La to Hg (d) Ac to Cn

(x) Oxidation of manganese in air gives the following oxide:
(a) MnO (b) MnO_3
(c) Mn_2O_3 (d) Mn_3O_4

Short Questions

1. Write the IUPAC names of the following:

(i) $Na_2[Pt(OH)_4]$

(ii) $K_2[Fe(CN)_5NO]$

(iii) $[Zn(NH_3)_4]^{+2}$

(iv) $[Ni(SCN)_4]^{-2}$

2. Give reasons for the following:

(i) Why do transition elements show variable oxidation states?
(ii) Why transition elements have ability to form alloys?
(iii) Why Cu^{+2} ion is blue but Zn^{+2} is colourless?

- Products that are made by using living organisms or their constituents such as enzymes, DNA etc are called products of biotechnology.
- The process in which coal is heated in the absence of air to produce coke, coltar, coal gases and ammonia liquour is known as destructive distillation of coal.
- Octance number measures the quality of fuel in terms of its capacity to with stand knocking during combustion.
- The C_{60} molecule of Bucky ball consists of twelve (12) five membered rings and twenty (20) six membered rings.
- Functional group is a specific group of atoms within the molecule which is responsible for unique chemical properties.
- A series of organic compounds in which the various members have similar structural features but differ from each other by methylene group ($-CH_2-$) is called homologous series.

Multiple Choice Questions

(i) The functional group that contains a nitrogen atom is:
(a) Alcohol (b) Ketone
(c) Ester (d) Amide

(ii) The compound belongs to the alkene family is:
(a) CH_4 (b) C_2H_6
(c) C_3H_8 (d) C_2H_4

(iii) Destructive distillation of coal gives coke, coaltar, coal gas and:
(a) Natural gas (b) Petroleum
(c) Naphtha (d) Ammonia liquor

(iv) The knocking of internal combustion engine can be reduced by the following petroleum process:
(a) Reforming (b) Refining
(c) Distillation (d) Condensation

(v) The general formula of the homologous series of alcohol is:
(a) C_nH_{2n} (b) C_nH_{2n+2}
(c) $C_nH_{2n+1}OH$ (d) $C_nH_{2n-1}OH$

(vi) The number of five membered and six membered rings in C_{60} Bucky ball are respectively:
(a) 12 and 12 (b) 5 and 15
(c) 12 and 20 (d) 40 and 20

(vii) The pair of compounds that exhibit isomerism is:
(a) C_2H_5-OH and C_3H_7-OH
(b) CH_3-CH_2-CH_2-CH_3 and CH_3-$CH(CH_3)$-CH_3
(c) C_2H_5-CH_2-Cl and C_3H_7-CH_2-Cl
(d) CH_3-NH_2 and CH_3-CH_2-NH_2

(viii) The type of coal that is hard and high ranked is:
(a) Peat (b) Lignite
(c) Bituminous (d) Anthracite

(ix) Which of the following hydrocarbon is the chief constituent of natural gas?
(a) CH_4 (b) C_2H_6
(c) C_3H_8 (d) C_4H_{10}

(x) Urea was first synthesized by Wohler from an inorganic material named as:
(a) Ammonium nitrate (b) Ammonium chloride
(c) Ammonium cyanate (d) Ammonium bicarbonate

Short Questions

1. Define organic chemistry and give the significance of some organic compounds in everyday life
2. What is Vital force theory? Why was it disapproved?
3. Define functional group and write the structure of three oxygen containing functional group.

- The prefix n, Iso and Neo are utilized for the common naming of alkanes where required.
- In IUPAC system the priority for the numbering of carbon chain on the basis of functional group and other substituents is given as.

  (-COOH) > (-CHO) > (C = O) > (-OH) > ($-NH_2$) > (-O-) > (=)> (≡)
- When writing the IUPAC name of organic compounds, first write the name of all prefixes, then name of the carbon chain and finally write the name of suffix.
- In IUPAC system, the suffix of alcohol is –ol, aldehyde is –al, ketone is –one carboxylic acid is –oic acid and ester is –oate.

Multiple Choice Questions

(i) Which of the following molecule possess acyl functional group:

(a) R-CO-X (b) R-$CONH_2$

(c) R-COO-CH_3 (d) R-CO-R

(ii) A hydrocarbon with the molecular formula C_7H_{12} is possibly:

(a) Heptane (b) Heptene

(c) Heptyne (d) Hepta diene

(iii) An organic compound possesses the structural formula $CH_3CH = CH – C \equiv CH$, its correct IUPAC name is:

(a) 1-pentyne-3-ene (b) 3-pentene-1-pentyne

(c) 3-penten-1-yne (d) 2-pentene-3-yne

(iv) IUPAC name of isopropyl alcohol is:

(a) 1-propanol (b) 2-propanol

(c) 1-butanol (d) 2-butanol

(v) Resorcinol is a phenol, it contains two –OH groups at:

(a) Position 1 and 2 (b) Position 1 and 3

(c) Position 1 and 4 (d) Position 2 and 4

(vi) Formula of a saturated hydrocarbon is C_4H_8, it should be:

(a) Butane (b) Butene

(c) Butyne (d) Cyclobutane

(vii) In the IUPAC naming of ketones the ending –e of main carbon chain is replaced by:

(a) yl (b) ol

(c) al (d) one

(viii) In the common system, carboxylic acid with six carbon atoms in straight chain is named as:

(a) propionic acid (b) valeric acid

(c) caproic acid (d) steric acid

(ix) The correct structure of 1, 3-penta diene is:

(a) $H_3C-CH=CH-HC=CH_2-CH_3$ (b) $CH_2=CH-CH_2-CH=CH_2$

(c) $CH_2=CH-HC=CH-CH_3$ (d) $CH_3-CH=C=CH-CH_3$

(x) The IUPAC name of $CH_3COOCH(CH_3)_2$ is:

(a) propyl ethanoate (b) ethyl propanoate

(c) isopropyl acetate (d) isopropyl ethanoate

Short Questions

1. Why IUPAC system of nomenclature is preferred on common system of naming of organic compounds?
2. Write the name of five organic families and mention the functional group present in them.
3. Write down basic rules for IUPAC naming of organic compound.
4. We often use the term Iso and Neo in the common system of naming of alkanes. Explain with example
5. Write the common names of first five members of carboxylic acid.
6. If an organic compound contains both double and triple bond, in the main carbon chain, what rules you follow to write its IUPAC name. Explain by giving an example.

EXERCISE

Multiple Choice Questions

(i) The final product obtained when hydrogen bromide (HBr) is added to an ethyne molecule:

(a) Bromo ethene
(b) 1, 1-dibromo ethane
(c) 1,2-bromo ethane
(d) 1,1,2,2 tetra bromoethane

(ii) The formula of a saturated hydrocarbon is C_3H_6, it should be:

(a) Propane
(b) Propene
(c) Propyne
(d) Cyclopropane

(iii) Ozonide on heating with zinc dust produce:

(a) Alcohol
(b) Aldehyde
(c) Alkene
(d) Ether

(iv) Which of the following pairs of compounds represent functional group isomerism?

(a) 1-butene and 2-butene
(b) Ethanol and dimethyl ether
(c) n-butane and iso butane
(d) Diethyl ketone and methyl propyl ketone

(v) The substituent that can act as a meta director is:

(a) -Cl
(b) $-CH_3$
(c) -OH
(d) -COOH

(vi) Welding gas among the following is:

(a) Ethylene
(b) Acetylene
(c) Ethane
(d) Methane

(vii) Benzene burns with smokey flame because of its:

(a) Inflammability
(b) High carbon % age
(c) High resonance energy
(d) Aromaticity

(viii) Select the suitable chemical to distinguish between Ethene and Ethyne:

(a) Alkaline $KMnO_4$
(b) Acidified $KMnO_4$
(c) Bromine water
(d) Ammonical $AgNO_3$

(ix) Meta directing group among the following is:

(a) – OH (b) – NH_2

(c) – CH_3 (d) – NO_2

(x) Acylation of benzene in the presence of $AlCl_3$ gives:

(a) Toluene (b) Acetophenone

(c) Phenol (d) Xylene

Short Questions

1. Give three differences between aliphatic and aromatic hydrocarbons?
2. Give the mechanism of free radical reaction between methane and chlorine in the presence of sunlight.
3. Complete the following reactions and name the major product formed in each reaction.

$$C_2H_5OH \xrightarrow{H_2SO_{4(conc.)}\ 170°C}$$

$$C_2H_5Br + KOH \xrightarrow{\text{Alcohol, heat}}$$

$$CH_3 - CH = CH_2 + HBr \longrightarrow$$

$$CH_2 = CH_2 + O_3 \longrightarrow$$

4. Define a chiral carbon? Give an example to justify your answer.
5. Write the equations with all required conditions for the reaction of benzene with the following:

(i) H_2 (ii) CH_3COCl (iii) $HNO_{3(conc.)}$ (iv) $H_2SO_{4(conc.)}$

6. Bring about the following conversions.
   (i) Toluene to ortho-para nitro benzoic acid
   (ii) Benzene to m-nitro toluene
7. How can you prepare ethene from dehydration of ethanol and ethyne from dehydro halogenations of ethyl chloride?
8. Alkanes are generally referred as parafins due to their less reactivity, why are they stable towards chemical reactions?

- Amines are derivatives of ammonia, formed by the replacement of one or more hydrogens by alkyl or aryl groups.
- Amines are classified as primary, secondary and tertiary depending upon the number of alkyl groups attached to nitrogen atom.
- Amines serve as bases; due to the presence of lone pair of electrons on nitrogen atom and have ability to accept proton from acids.
- Amines are prepared by alkylation of ammonia, also prepared by reduction of nitriles and amides.

Multiple Choice Questions

(i) Which of the following composition justifies the secondary alkyl halide?
(a) R_3CX (b) R_2CHX
(c) RCH_2X (d) CH_3X

(ii) Which of the following alkyl halide cannot produce an alkene while treated with alcoholic potassium hydroxide:
(a) Methyl bromide (b) Ethyl bromide
(c) Propyl bromide (d) Butyl bromide

(iii) Ethyl magnesium bromide with carbon dioxide yields.
(a) Methanoic acid (b) Ethanoic acid
(c) Propanoic acid (d) Butanoic acid

(iv) Grignard reagent with ester produces:
(a) Aldehyde (b) Carboxylic acid
(c) Ketone (d) Ether

(v) Amines act as bases because:
(a) They accept OH^- (b) They accept H^+
(c) They donate H^+ (d) They donate the OH^-

(vi) The structure of Primary amine is:
(a) Planar trigonal (b) Linear
(c) Tetrahedral pyramidal (d) Regular tetrahedral

(vii) Alkyl amine when reacts with nitrous acid in the presence of hydrochloric acid, yields:

(a) Diazonium salt (b) Aldehyde
(c) Ketone (d) Alcohol

(viii) SN^2 reaction occurs most easily if the substrate molecule is:

(a) A methyl iodide (b) An ethyl iodide
(c) 2-iodo propane (d) 2-iodo butane

(ix) Suitable reagent required for the synthesis of propane from methyl magnesium iodide is:

(a) H_2O (b) NH_3
(c) CH_3OH (d) CH_3NH_2

(x) The rate of SN^1 mechanism depends upon:

(a) Conc. of substrate only
(b) Conc. of attacking nucleophile only
(c) Conc. of both substrate and attacking nucleophile
(d) Polar solvent

Short Questions

1. How are alkyl halides prepared by the reaction of alcohol with (i) HX (ii) PX_3 (iii) $SOCl_2$. Give the equations.
2. How are amines prepared from nitriles, give the equations.
3. Why are secondary and tertiary amines more alkaline than primary amines?
4. Why the alkyl part of Grignard reagent is nucleophilic in nature?
5. Convert the followings:
   (a) Methyl magnesium bromide into acetone
   (b) Ethylchloride into ethyl amine
   (c) Ethyl chloride into ethyl alcohol
   (d) Ethylamine into imine
6. How is primary amine converted into secondary and tertiary amines, give the equations.
7. Give an account on the basicity of amines.
8. How can you justify the fact that alkyl halides are water insoluble?
9. How can you define a nucleophile? Write the names of four nucleophiles along with their typical reagents.
10. Why the mechanism of SN^2 reaction completes in one step?

- Acidic character of phenol (pKa = 10) is higher than alcohol (pKa ≈ 16) because phenoxide ion is more stable than alkoxide ion.
- Boiling points of alcohol and phenol are higher than alkanes and ethers of comparable molecular mass because they form intermolecular hydrogen bonds.
- Both alcohols and phenols are water soluble but solubility of alcohols is higher than phenols.
- The cleavage of ethylene glycol in the presence of periodic acid (HIO_4) gives two formaldehyde molecules.
- Phenols are commercially prepared by Dow's Process during which chlorobenzene is fused with aqueous sodium hydroxide at 350°C and 150 atmospheric Pressure.
- The aromatic part of phenol is similar to the benzene which favours electrophilic substitution reaction.
- Phenols are identified by bromine water test as well as ferric chloride test.
- Alcohols are identified by sodium metal test and ester test.
- Ethers are considered as the dialkyl or diaryl derivatives of water. They are represented by a general formula R-O-R.
- Boiling point of ethers are much lower than alcohol and phenol since they do not form intermolecular hydrogen bonds.
- Symmetrical ethers are those in which similar alkyl or aryl groups are attached with oxygen atom.
- Ethers are relatively less reactive because C-O-C bond of ether is susceptible for nucleophilic or electrophilic attack.

Multiple Choice Questions

(i) In the molecule of phenol, the carbon atom which is attached to hydroxyl group is.

(a) sp-hybridized (b) sp^2-hybridized
(c) sp^3-hybridized (d) Unhybridized

(ii) Which of the following is a trihydric phenol?

(a) Resorcinol (b) Cresol
(c) Pyrogallol (d) Catechol

(iii) Ethanol reacts with PCl_3 to form.

(a) Diethyl ether (b) Ethene
(c) Ethyl chloride (d) Ethanoic acid

(iv) Which of the following alcohols has highest boiling Point

(a) Ethyl alcohol (b) n-pentyl alcohol
(c) iso-pentyl alcohol (d) neo-pentyl alcohol

(v) Which of the following Products is mainly formed if ethanol is dehydrated with concentrated sulphuric acid at 170°C?

(a) Ethene (b) Ethyne
(c) Ethanol (d) Diethyl ether

(vi) Lucas reagent is a mixture of

(a) Zn and Hg (b) Zn and HCl
(c) $ZnCl_2$ and HCl (d) NaOH and CaO

(vii) Oxidative cleavage of 1, 2 – diol occur in the presence of

(a) $K_2Cr_2O_7$ (b) $KMnO_4$
(c) HNO_3 (d) HIO_4

(viii) Which of the following molecule cannot form hydrogen bonding with water molecule?

(a) Phenol (b) Resorcinol
(c) Ethyl chloride (d) Ethyl alcohol

(ix) Secondary alcohols, undergo oxidation with potassium dichromate to produce carboxylic acid through an intermediate product known as:

(a) Aldehyde (b) Ketone
(c) Ether (d) Alkyl halide

(x) Which of the following is an anaesthetic agent

(a) Phenol (b) Ethyl alcohol
(c) Diethyl ether (d) Acetone

Short Questions

1. Define Phenol? Write the equations for the preparation of Phenol from.
   (i) Chlorobenzene
   (ii) Sodium benzene sulphonate
2. Write the equations for the following chemical Process
   (i) Reduction of acetic acid with $LiAlH_4$.

- Ketones can be oxidized into carboxylic acid in presence of strong oxidizing agent.
- Nucleophilic addition reactions of aldehydes and ketones are catalysed by acid or base depending upon the availability of either weak or strong nucleophilic reagent.
- Water, alcohol, and ammonia can undergo nucleophilic addition reactions with aldehydes and ketones, and these reactions are catalyzed by acids.
- Hydrogen cyanide (HCN) and Grignard reagent (RMgX) can react with aldehydes and ketones in the presence of a base catalyst.
- Identification of aldehydes in a laboratory can be carried out by Tollen's reagents or by Fehling solution.

Multiple Choice Questions

(i) Ketones when treated with $LiAlH_4$, they reduce to:
(a) Primary alcohol (b) Secondary alcohol
(c) Tertiary alcohol (d) Dihyderic alcohol

(ii) The reagent used to oxidizes ketones into carboxylic acids is:
(a) Ammoniacal silver nitrate (b) Potassium dichromatic
(c) Fehling solution (d) Benedict solution

(iii) The carbonyl carbon of aldehydes and ketones is:
(a) Sp hyrbidized (b) Sp^2 hyrbidized
(c) Sp^3 hyrbidized (d) dsp^3 hyrbidized

(iv) Acetophenone is the member of ketone family, it contains:
(a) Two alkyl groups (b) Two aryl group
(c) One alkyl and one aryl group (d) One aryl and one hydrogen atom

(v) The most reactive molecule towards nucleophilic addition in the following is:
(a) Formaldehyde (b) Acetaldehyde
(c) Diethyl ketone (d) Acetophenone

(vi) Clemmensen reduction is the conversion of aldehydes and ketones into:
(a) Alkanes (b) Alkenes

(c) Alkyl halides (d) Alcohols

(vii) Hydration of propyne in the presence of H_2SO_4 and $HgSO_4$ gives:
(a) Formaldehyde (b) Methyl ethyl ketone
(c) Acetone (d) Acetaldehyde

(viii) Which of the following carbonyl compound is most soluble in water?
(a) Formaldehyde (b) Acetaldehyde
(c) Benzaldehyde (d) Acetophenone

(ix) Which of the following gives silver test with Tollen's reagent?
(a) HCHO (b) CH_3-O-CH_3
(c) C_2H_5OH (d) CH_3COOH

(x) On reduction of a carbonyl compound by Zn-Hg and Conc. HCl, it is converted to an alkane. This reaction is known as;
(a) Dow reduction (b) Cope reduction
(c) Clemmensen reduction (d) Wolf-Kishner reduction

Short Questions

1. Give reasons for the following:
   i. The boiling point of aldehydes and ketones is lower than alcohol.
   ii. Formaldehyde is highly soluble in water as compared to other aldehydes.
   iii. Oxidation of aldehydes is faster than ketones.
2. Write the equation for the reaction of acetaldehyde with the following:
   i. Chromic acid (H_2CrO_4)
   ii. Lithium Aluminum hydride ($LiAlH_4$)
   iii. Zinc mercury amalgam
3. How is formaldehyde prepared by ozonolysis?
4. Why is formaldehyde more reactive towards Nucleophilic addition reactions compared with ketones.
5. How does the oxidation of ketones differ from the oxidation of aldehydes?
6. Why are ethers considered as good solvents in organic reactions?

Descriptive Questions

1. What are aldehydes and ketones? Describe the structure and type of hybridization in them.
2. Explain the acid-catalyzed and base-catalyzed nucleophilic addition reactions in aldehydes and ketones.

- Carboxylic acids can be converted into acyl chloride, if they are treated with PCl_3, PCl_5 or $SOCl_2$.
- Carboxylic acids can be converted into esters, if they are heated with an alcohol in the presence of concentrated sulphuric acid.
- Carboxylic acids can be converted into amide, if they are treated with ammonia.

Multiple Choice Questions

(i) The most common compound found in pineapple is:
(a) Acetic acid (b) Ethanol
(c) Acetone (d) Ethyl butanoate

(ii) Two molecules of acetic acid on condensation gives:
(a) Ethyl acetate (b) Aceticamide
(c) Acylhalide (d) Acetic anhydride

(iii) Carboxylic acid is stronger acid than:
(a) HCl (b) HNO_3
(c) C_2H_5OH (d) H_2SO_4

(iv) The reagent that cannot produce an acyl halide in reaction with a carboxylic acid is:
(a) PCl_3 (b) PCl_5
(c) HCl (d) $SOCl_2$

(v) Benzoic acid is the product of oxidation of:
(a) Benzene (b) Ethyl Benzene
(c) Aniline (d) Phenol

(vi) Formation of acyl halide and amide by carboxylic acid involved
(a) Replacement of hydrogen (b) Replacement of carbonyl group
(c) Replacement of hydroxyl group (d) Replacement of oxygen

(vii) Formic acid is naturally found in:

(a) Venom of ants (b) Bees string

(c) Vinegar (d) Butter

(viii) Among the following compounds, the one with the highest boiling point is:

(a) Ethanol (b) Acetaldehyde

(c) Acetic acid (d) Ethyl chloride

(ix) The formula of caproic acid is:

(a) C_4H_9COOH (b) $C_5H_{11}COOH$

(c) $C_6H_{13}COOH$ (d) $C_7H_{15}COOH$

(x) The reaction of acetic acid with ethanol in the presence of conc. sulphuric acid gives:

(a) Ethyl acetate (b) Acetamide

(c) Ethane (d) Acetic anhydride

Short Questions

1. Explain why?
   i) The boiling points of carboxylic acids are high than alcohol?
   ii) The structure of carboxylic acid is trigonal planar?
2. What happens when:
   i) Formaldehyde reacts with a mixture of Potassium dichromate & sulphuric acid.
   ii) Carboxylic acid reacts with thionyl chloride
   iii) Carboxylic acid reacts with ammonia
   iv) Ethyl magnesium bromide reacts with carbon dioxide
3. Write down the commercial applications of carboxylic acids.
4. Write the natural sources of following carboxylic acids.
   (a) Formic acid (b) Acetic acid (c) Valeric acid (d) Caproic acid

- Lipids are water insoluble heterogeneous organic compounds, which are chemically ester of fatty acid and alcohols.
- Meat, nuts, cereals, fish, milk and dairy foods, fruits and vegetables, are sources of minerals.
- Saponification is a chemical reaction of fat and oil with an alkali.
- Rancidity involves two simultaneous reactions named as oxidation and hydrolysis which makes the animal fat bad taste and bad odour.

Multiple Choice Questions

(i) Starch and Sucrose are examples of:
(a) Monosaccharides and Disaccharides
(b) Disaccharides and Oligosaccharides
(c) Polysaccharides and Disaccharides
(d) Monosaccharides and Polysaccharides

(ii) Amino acid units bonded in protein molecule through:
(a) Glycosidic linkage (b) Ether linkage
(c) Peptide linkage (d) Hydrogen bridge

(iii) Proteins are composed of:
(a) Amino acids (b) Carbohydrates
(c) Lipids (d) Nucleic acids

(iv) A condensation polymer of amino acid is:
(a) Protein (b) Lipids
(c) Starch (d) Glycogen

(v) Saponification is the formation of soap by the reaction of fat and oil with:
(a) An alkali (b) An acid
(c) Sugar (d) Glycerol

(vi) Which of the following mineral is considered to be essential for immune system:

(a) Iron (b) Zinc
(c) Magnesium (d) Calcium

(vii) Rancidity is a chemical process involving:

(a) Oxidation and hydrolysis (b) Condensation and reduction
(c) Polymerization (d) Decarboxylation

(viii) Lipid which is a major component of cell membrane is:

(a)Triglyceride (b) Phospholipid
(c) Glycolipid (d) Steroid

(ix) Total numbers of alpha amino acids are:

(a) 19 (b) 22
(c) 25 (d) 28

(x) Sugar molecules are classified as:

(a) Fats (b) Proteins
(c) Carbohydrates (d) Lipids

Short Questions

1. Mention the three main functions of lipids.
2. Comparing with other nutrients, why lipids are better source of energy?
3. Carbohydrates are necessary component of our diet. Give two dietary importance of carbohydrates.
4. What is meant by saponification? Give the reaction.
5. What is rancidity which chemical reaction involves in this process?
6. Write three essential functions of protein in the body.
7. Write down the sources from which we intake fructose and lactose.

EXERCISE

Multiple Choice Questions

(i) Super glue is chemically named as:
(a) Cyano acrylate (b) Polyvinyl acetate
(c) Epoxy resins (d) Polyurethane

(ii) DDT is a chemical which commercially known as:
(a) Insecticide (b) Herbicide
(c) Pesticide (d) Fungicide

(iii) Nail polish remover is mainly consists of:
(a) Pigments (b) Acetone
(c) Diethyl ether (d) Ethyl alcohol

(iv) Antimalarial drug among the following is:
(a) Ibuprofin (b) Chloroquine
(c) Paracetamol (d) Diphenyl hdramine

(v) Asprin is a pain reliever, its chemical name is:
(a) Ascorbic acid (b) Nicotinic acid
(c) Acetyl salisylic acid (d) Benzoic acid

(vi) Nylon 6, 6 is a condensation polymer of hexamethylene diamine and:
(a) Benzoic acid (b) Adipic acid
(c) Pthalic acid (d) Valeric acid

(vii) Which of the following is not a synthetic plastic:
(a) Nylon (b) Teflon
(c) Cellulose (d) Polyethene

(viii) Drugs that lower the body temperature to normal are known as:
(a) Antibiotics (b) Antipyretic
(c) Antiallergic (d) Anti histamins

(ix) Which of the following chemical is used as an oxidizing agent in permanent hair dyes:
(a) Acetone (b) Hydrogen peroxide
(c) Polyvinyl acetate (d) Resorcinol

(x) An example of thermosetting plastic is:
(a) Polyethene (b) PVC
(c) Nylon (d) Bakelite

EXERCISE

Multiple Choice Questions

(i) The greenhouse gas is:

(a) Oxygen (O_2) (b) Nitrogen (N_2)

(c) Argon (Ar) (d) Carbon dioxide (CO_2)

(ii) The pH of acid rain is:

(a) Between 7 to 8 (b) Between 6 to 7

(c) Below 5 (d) Above 8

(iii) Ozone depletion in upper atmosphere is mainly caused by:

(a) Sulphur dioxide (SO_2) (b) Nitrogen oxides (NO_x)

(c) Carbon monoxide (CO) (d) Chlorofluorocarbons ($CFCs$)

(iv) The region of sphere which extend from 11km to 50km from our Earth's is known as:

(a) Troposphere (b) Stratosphere

(c) Mesosphere (d) Thermosphere

(v) The increase in global average temperatures is primarily attributed to:

(a) Solar radiations (b) Oxides of sulphur (SOx)

(c) Green house gases emission (d) Natural climate variations

(vi) The primary goal of green chemistry is:

(a) Maximizing industrial production

(b) Minimizing waste generation

(c) Reducing energy consumption

(d) Developing environmentally friendly chemicals

(vii) Smog is word used for the combination of:

(a) Water vapours and frog (b) Smoke and fog

(c) Sediments and colloids (d) Oxides of nitrogen and sulphur

(viii) The catalyst used in automobile catalytic converter is a mixture of:

(a) Pt and Pd (b) Ni and Fe

(c) Cu and Cr (d) Pb and Hg

(ix) Ozone layer is present in which of the following region of atmosphere:

(a) Troposphere (b) Stratosphere

(c) Mesosphere (d) Thermosphere

(x) The major source of water pollution is:

(a) Organic farming (b) Global warming

(c) Industrial activities (d) Renewable energy sources

Short Questions

1. Write the name of four segments of atmosphere and mention in which segment we live.
2. Explain the causes of depletion of ozone layer.
3. What is industrial smog and how is it formed?
4. What are the main gases responsible for green house effect?
5. Explain four fundamental methods for the testing of waste water.

Descriptive Questions

1. What is the effect of acid rain on human health and what measures can be taken to prevent acid rain?
2. What is the main cause of Global warming? How does it effect on weather pattern?
3. Describe the chemistry involves due to the presence of oxides of nitrogen and sulphur in the troposphere.
4. What is Green house effect? How does human activities contributes to the enhancement of the green house effect?
5. What are ozone depletion substances (ODS)? What human activities have contributed to ozone depletion in the stratosphere?
6. What is a catalytic converter? What are the main pollution targeted by catalytic converter?

EXERCISE

Multiple Choice Questions

(i) The energy of IR radiations is:
(a) Higher than radio waves
(b) Higher than U.V waves
(c) Lower than micro waves
(d) Higher than x-rays

(ii) The fingerprint region of an IR spectrum typically found in the range of:
(a) 1500-4000cm^{-1}
(b) 2000-4000cm^{-1}
(c) 1500-500cm^{-1}
(d) 4000-8000cm^{-1}

(iii) Which unit of measurement is used for the wave number of IR spectrum:
(a) Nanometer (nm)
(b) Angstrom (Å)
(c) Centimeter^{-1} (cm^{-1})
(d) Hertz (Hz)

(iv) Electronic excitation occurs in electromagnetic spectrum if the molecule absorbs:
(a) I.R radiations
(b) U.V-visible radiations
(c) Radio waves
(d) Micro waves

(v) In NMR spectroscopy, the hydroxyl proton of C_2H_5OH appears as a broad singlet around:
(a) 1 – 2 ppm
(b) 2 – 3 ppm
(c) 4 – 5 ppm
(d) 6 – 7 ppm

(vi) Atomic absorption spectrum is represented by:
(a) Dark lines against bright background
(b) Bright lines against dark background
(c) Bright lines against bright background
(d) Dark lines against dark background

(vii) Infrared spectroscopy is a technique use to determine _____ in the given organic molecule:
(a) Double and triple bonds
(b) Mass to charge ratio
(c) Functional group
(d) Conjugated system

(viii) Highest U.V-visible absorption energy require for the transition of:

(a) σ to σ* (b) π to π*

(c) n to σ* (d) n to π*

(ix) NMR spectroscopy is carried out if radiations interact with molecules in high magnetic field:

(a) U.V-visible rays (b) Radio rays

(c) X-rays (d) Infrared rays

(x) In which of the following spectroscopy technique D_2O can be used as solvent:

(a) Mass spectroscopy (b) NMR spectroscopy

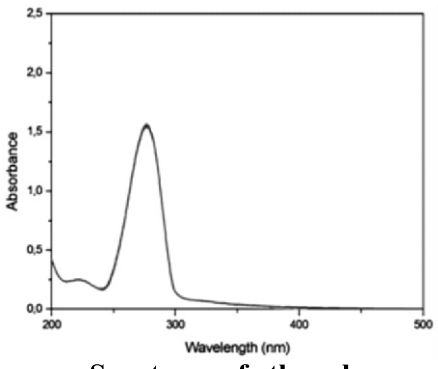
(c) Atomic absorption spectroscopy (d) I.R spectroscopy

Short Questions

1. What types of nuclei are detected in proton NMR spectroscopy?
2. Name the components which represents x-axis and y-axis of a proton NMR spectrum.
3. Differentiate between atomic absorption and emission spectroscopy.
4. What is the purpose of U.V-visible spectroscopy? What is its applications in chemistry and biology?

Descriptive Questions

1. What information about the structure of a molecule we can get from mass spectroscopy? Give the applications of mass spectroscopy.
2. What is proton NMR spectroscopy? How does it work? Give its applications.
3. Explain λ_{max} with the help of U.V-visible spectrum of ethanol (C_2H_5OH).



Spectrum of ethanol